Towards Full Understanding of Stress using Machine Learning and Multiple Sensors

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ABSTRACT

This paper explores physiological responses to stress among college students through wearable sensor data and machine learning techniques. Four students wore an Actigraph watch and Hexoskin vest during classes over two months, capturing movement, cardiovascular, and respiratory data. Participants self-reported stress levels for each class. Exploratory analysis revealed increases in average heart rate aligned with heightened stress but no definitive physiologic differences between STEM and non-STEM classes. Models utilizing solely Actigraph data achieved only 48% accuracy in predicting stress levels, demonstrating limitations as standalone stress detectors. However, a model employing three key Hexoskin features (breathing volume, rate, and heart rate) attained 89.5% accuracy, rivaling more complex methodologies requiring invasive equipment. Breathing volume emerged as the predominant predictor, despite minimal coverage in existing literature. These findings reveal feasibility for an affordable, discreet vest measuring said metrics to unobtrusively detect stress. This could enable real-time notifications prompting tension-alleviating interventions, thereby improving student well-being, retention, and academic performance. In summary, the research pioneers an interpretable stress detection framework with implications for supporting students through emerging sensor technologies.

KEYWORDS

Stress Detection, Wearable Sensors, Machine Learning, Heart Rate Variability, Students, Hexoskin, K-Nearest-Neighbors, XGBoost, Affordability

1 INTRODUCTION

Chronic stress, a pervasive concern among college-age students, is notably linked to the academic environment and has been identified as a significant predictor of elevated stress levels and chronic stress in university students [11]. Recognizing the imperative nature of accurately identifying stressors, this study endeavors to comprehensively understand the detrimental effects of stress on college students [11]. This awareness is paramount given the prevalence of debilitating stress among students, emphasizing the importance of understanding these stressors as an initial step towards improving classroom performance and overall well-being. In contributing to the existing body of literature on stress detection, this study seeks to leverage machine learning models and data science methodologies.

The exploration encompasses various facets of stress comprehension, intending to expand our current knowledge of stress inducers. A crucial objective is the establishment of a framework for prioritizing biometric stressors, discerning the most influential contributors. Significantly, students reporting stress-related impacts on performance demonstrate lower GPAs, heightened stress levels, and diminished coping self-efficacy, resilience, and social support [9]. Consequently, the identification of biometric cues holds promise for developing strategies to mitigate stress, thereby potentially enhancing classroom performance and overall well-being. Additionally, the study seeks to investigate the association between class type (STEM versus NON-STEM) and stress responses.

Despite the acknowledged importance of stress detection, a challenge remains in the accessibility of sensors suitable for consumer use. Existing stress detection models may not be sufficiently advanced for commercialization, and college students often lack the financial resources and expertise for stress detection using sophisticated sensors. Addressing this gap, our study aims to evaluate the feasibility of a stress-prediction model based solely on movement data and its potential adaptation to affordable wearable watches. In the event that this approach proves impractical, the study will explore which subset of biometric features from the Actigraph GT9X Wearable Sensor Watch and the Hexoskin Vest Sensor are the most effective predictors of stress. This multifaceted investigation aims to provide valuable insights into stress detection methodologies that are both effective and accessible for college students.

2 RELATED WORK

The field of stress detection through machine learning and sensor technologies has witnessed extensive exploration, as evidenced by a robust literature encompassing various facets of this research. Current stress detection sensors, including Heart Rate Variability (HRV), Electrodermal Activity (EDA), Electromyography (EMG), Electrocardiogram (ECG), Galvanic Skin Response (GSR), Photoplethysmography (PPG), and Autonomic Nervous System Sensors (ANS), have been subject to comprehensive investigation in the existing literature [3, 8, 13]. However, these sensors often present challenges, either being too obtrusive for regular wear or awaiting widespread adoption among the general consumer base. The accuracy of these sensors ranged from 50% to 99%, with a general trend indicating that more accurate sensors tended to be more expensive or intrusive in nature.

The significance of HRV, derived from electrocardiography, as an indicator of stress and health has been well-documented. Wearable sensors, owing to their non-invasive nature, facilitate continuous monitoring, enabling machine learning techniques such as random forests to leverage HRV features for stress level estimation [7]. Median heart rate, in particular, emerges as a robust non-invasive predictor of stress [7]. Long-term monitoring through wearables,

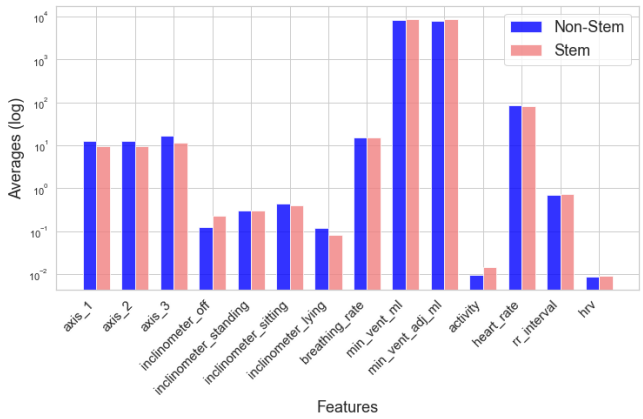

Figure 1: Average Differences in Feature Means (STEM Vs. NON-STEM Classes

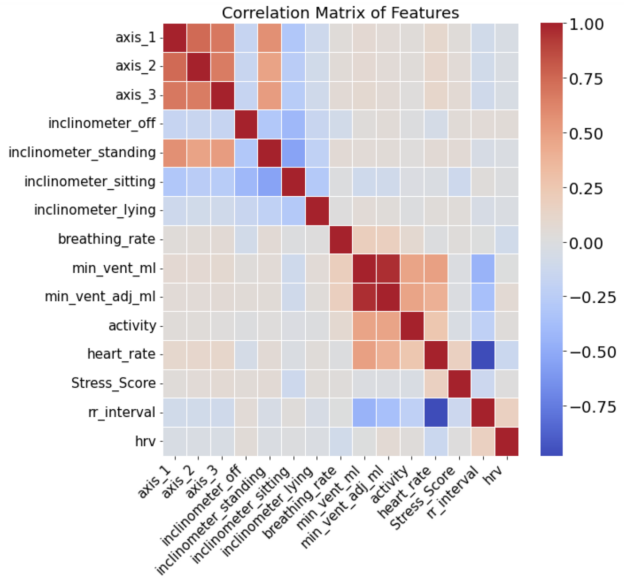

Figure 2: Correlation Matrix All Features and Stress

incorporating variables like heart rate, skin temperature, and self-reported stress, exhibits promise in predicting chronic stress over extended durations [4]. Literature reviews also recognize the potential of utilizing biomedical signals from wearables to understand indoor environmental impacts on comfort, health, and building performance [10].

This research assumes particular relevance for students, considering that psychological factors are identified as significant predictors of perceived stress [12]. Hybrid machine learning models, integrating physiological inputs, have demonstrated efficacy in predicting stress levels among students [1]. Moreover, the detection of fidgeting through wearables during lectures introduces a novel non-invasive indicator of stress and its potential influence on memory retention [2]. The integration of machine learning models leveraging physiological signals from wearables and movement data holds promise for the passive and accurate detection of stress in students, offering insights for targeted interventions to enhance learning environments. In summary, this research not only explores traditional stress detection sensors but also underscores the potential of wearables and machine learning models for comprehensive stress monitoring, with the aim of identifying features not currently addressed by existing methodologies and shedding light on specific stress responses in students.

3 STUDY AREA AND MATERIALS

3.1 Study Area and Time Period

This research is centered on the College of Charleston, a higher education institution characterized by a diverse student population engaged in various academic disciplines. The study involves a sample comprising two male and two female students at the Junior to Senior status levels, ensuring representation across diverse academic domains. The investigation spans multiple classes over a duration of 2 months. The selected classes for data collection are enumerated below, with corresponding annotations denoting whether they fall under STEM or NON-STEM classifications. The courses include Psychology 213, Psychology 315, Psychology 224, Geology 275, Geology 275 Lab, Computer Science 380, Computer Science 310, and Mathematics 440 (all STEM); and History 361, History 250, Honors Politics, Classics 200, and Politics 250, and Theatre 175 (all NON-STEM). This diverse selection of classes ensures a comprehensive representation of academic disciplines, allowing for a nuanced exploration of stress responses across STEM and NON-STEM contexts. The chosen timeframe of 2 months provides a sufficiently robust dataset for the study's objectives.

3.2 Actigraph Watch Data

The Actigraph GT9X wearable device provides triaxial accelerometer data, capturing instantaneous acceleration along the x, y, and z axes, expressed in gravitational units (Gs). Additionally, the device is equipped with an inclinometer, facilitating the detection of various body orientations, including standing, lying, sitting, and off. Data acquisition occurs at a resolution of 60Hz, resulting in one data point per second. The Accelerometer (Axis 1, 2, 3), provides instantaneous acceleration in Gs with a dynamic range of +/- 16Gs per axis, and the Inclinometer (Off, Lying, Sitting, Standing), captures diverse body orientations. The Actigraph feature names are axis_1, axis_2, axis_3, inclinometer_off, inclinometer_lying, inclinometer_sitting, and inclinometer_standing. These features enable a detailed analysis of movement patterns and body orientations, contributing to the comprehensive assessment of stress responses among college students.

3.3 Hexoskin Vest Data

The Hexoskin vest serves as a sophisticated physiological monitoring tool, capturing essential parameters related to respiratory and cardiovascular activity. Operating at a 60Hz resolution, this vest offers a granular perspective into physiological dynamics, collecting data at a frequency of 1 time per second. Breathing Rate (calculated over the last 7 respiration cycles with a valid range of 2 to 60 RPM); Heart Rate (average over the last 16 heartbeats within a valid range

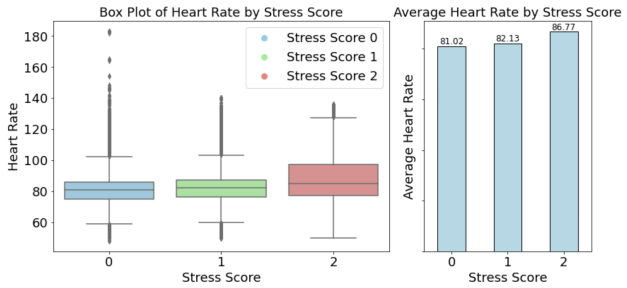

Figure 3: Heart Rate Distribution and Averages per Stress Score

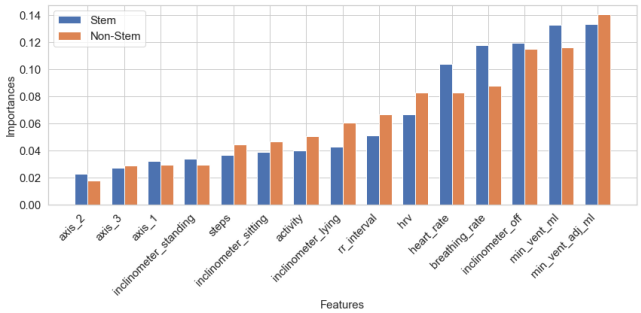

Figure 4: Feature Importance Using Hybrid Model (STEM Versus NON-STEM)

of 30 to 220 BPM); Minute Ventilation (representing the volume of air per minute inspired in the last 7 respiration cycles); Minute Ventilation (Adjusted, accounting for user profile adjustments); and Activity (a signal derived from accelerometer intensity vector with a high-pass filter at 2.65 Hz). The Hexoskin feature names are breathing_rate, heart_rate, min_vent_ml, min_vent_adj_ml, and activity. These features facilitate a nuanced exploration of respiratory and cardiovascular dynamics, contributing to a comprehensive understanding of physiological responses associated with stress among college students.

4 METHODS

4.1 Data Collection

The data collection process involves the initialization of both the Actigraph watch and the Hexoskin vest to coincide with each participant's daily class schedule. Participants are required to wear the Hexoskin vest throughout the day, initiating and concluding activity recording within the Hexoskin app for each class. Simultaneously, participants wear the Actigraph watch during their scheduled classes, with data collection occurring automatically based on predefined start and stop times, requiring no additional user input. This approach aims to streamline the data collection process while ensuring accurate synchronization with the participants' class schedules. At the end of each class participants are instructed to rate their stress score for each class that day: 0 being no stress, 1 being mild stress, and 2 being high stress.

A noteworthy aspect of the data collection methodology is the adoption of a per-activity basis for gauging stress scores, as opposed to a singular, all-encompassing stress score. This approach is supported by findings in the literature, emphasizing the increased accuracy associated with assessing stress on a per-activity basis [6]. The duration of class periods varied, ranging from 1 hour and 15 minutes to 3 hours, with the majority falling within the one-hour and 15-minute timeframe. This diverse class duration distribution provides a comprehensive dataset for the study's objectives.

4.2 Actigraph Watch Data Preprocessing

The Actigraph watch data underwent minimal preprocessing, primarily owing to the predefined time intervals specified before the commencement of data collection. As a result, there was no need for extensive preprocessing steps. During Actigraph data collection, exclusion criteria were applied to remove data points corresponding to periods when the student was not actively engaged in one of their scheduled classes, as outlined in their class schedule.Null values in the Actigraph watch data were encountered due to inherent sensor characteristics. However, the prevalence of null values was limited, ranging from 5 to 10 instances per data collection use. The handling of these null values was addressed to maintain data integrity and ensure the reliability of subsequent analyses.

4.3 Hexoskin Vest Data Preprocessing

Preprocessing of the Hexoskin vest data involved the trimming of small time intervals, typically in the few-minute range, to address anomalies such as classes ending earlier than scheduled. Null values were identified in the Hexoskin vest data, stemming from inherent sensor characteristics. However, the incidence of null values remained relatively low, ranging from 5 to 10 instances per data collection use. To maintain data integrity and facilitate subsequent analyses, these null values were systematically removed from the dataset. This approach ensured the reliability of the Hexoskin vest data for the study's overarching objectives.

4.4 Data Combination and Final Preprocessing

To amalgamate data from both wearable devices, synchronization is performed based on their starting times, with adjustments made to the smaller dataset to align with the temporal scope of the larger one. Outliers are treated independently for each dataset.

As outlined in the study protocol, participants assign stress ratings at the conclusion of each day for every class, utilizing a scale ranging from 0 (indicating no stress) to 2 (indicating high stress), with an intermediate value of 1 signifying mild stress. These stress scores are subsequently appended to the combined data frames (vest and watch) in accordance with the daily format of the study by Hosseini et al. [4].

Incorporating a physiological marker of stress, Heart Rate Variability (HRV) is computed based on established methodologies from prior studies [5]. HRV is derived from the heart rate and time columns using a rolling window of 5 seconds. The computation involves dividing the range of heart rate values within each 5-second window by the standard deviation of those values. Subsequently, any unpopulated cells in the dataset are filled through an iterative process based on previously calculated HRV values.

4.5 Additional Features Description

In addition to stress-related metrics, two supplementary physiological features are introduced. Heart Rate Variability (HRV): Computed through a rolling window of 5 rows based on the heart rate column. The calculation involves dividing the range of heart rate values within each window by the standard deviation of those values. Any vacant cells in the dataset are then populated with values derived from preceding HRV computations. RR Interval: Calculated by identifying consecutive peaks in the heart rate data, representing the time intervals between consecutive heartbeats. The designations assigned to the supplementary features incorporated in the dataset are as follows are hr and rr_interval.

4.6 Data Exploration

A focal point of interest lies in discerning the relationship between biometric predictors of stress and individual students' experiences across distinct class types (STEM or Non-STEM).

Upon scrutinizing the mean feature levels within STEM and Non-STEM datasets for all individuals across stress scores (0, 1, and 2), no discernible physiological pattern emerges in association with stress and class type. As depicted in the figure (see Fig. 1). The average feature responses for low, medium, and high stress (corresponding to stress scores 0, 1, and 2) exhibit negligible differences. The figure illustrates averages across all stress scores due to the striking similarity in features associated with individual stress scores. However, STEM students tend to manifest slightly higher average stress scores, a trend possibly attributed to the disproportionate data collection from STEM classes as opposed to non-STEM classes.

Furthermore, a consistent finding indicates an overall increase in heart rate across all participants and classes with ascending stress scores (0, 1, and 2), corroborating the findings of a related study [7] (see Fig. 3).

The congruence between the collected data and the outcomes of the cited study, particularly in terms of heart rate, underscores the validity of the acquired dataset. Notably, an observable escalation in average heart rate coincides with heightened stress levels.

In summary, while no substantial disparity is evident in the physiological responses between STEM and Non-STEM classes, a consistent elevation in heart rate is discernible.

Subsequently, an exploration was conducted to ascertain the feasibility of constructing a stress-detection model using solely movement data from the watch. Correlations between actigraph and hexoskin features were examined in an attempt to predict stress levels exclusively from the watch. However, no significant correlations were discerned in these features, suggesting limited insights into the predictive capacity of actigraph features regarding stress levels or scores (see Fig. 2).

4.7 Model Construction and Further Data Exploration

The aim was to predict the appended stress scores with different classifications model. The process of model construction commenced with a 5-fold stratified cross-validation, a well-established practice in the literature [1, 8]. The choice of 5 or 10 folds aligns

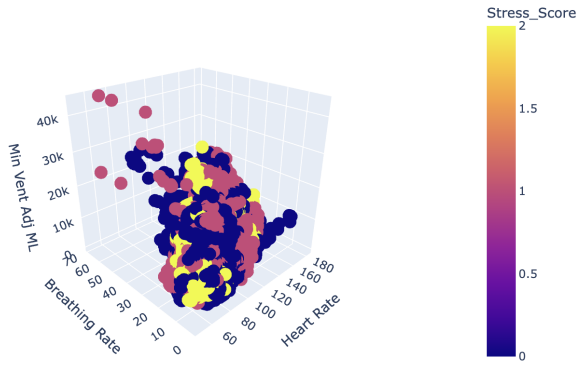

Figure 5: Stress Scores for top 3 features

with recognized standards [1, 8]. A predetermined seed was employed to facilitate the comparison of model performance across different features and models.

During model testing, the dataset incorporated all participants along with their respective classes, providing a diverse testing environment and resembling a representative sample of the general population, as observed in a pertinent study [8]. It is noteworthy that the model was trained on data from four participants and their classes, testing randomly on specific participants' data.

Further testing involved the division of all participants' data into stem and non-stem dataframes. This approach allowed for an exploration of model performance within these specific subsets, contributing to a nuanced understanding of potential distinctions.

Three distinct models—Random Forest, XGBoost, and a Random Forest-XGBoost Hybrid Model—were employed. The selection of these models was guided by existing literature citing their superior performance, albeit at the expense of interpretability regarding specific feature contributions values to stress scores [1, 6, 7]. Additionally, the computational efficiency of the models was considered, with the random forest model exhibiting a runtime of 5-10 minutes, prompting the exploration of faster alternatives.

A grid search was conducted to fine-tune the models, specifically determining the optimal number of estimators for the random forest and XGBoost models. The identified values were 100 for the random forest model and 300 for the XGBoost model, respectively.

In an effort to amalgamate insights from both XGBoost and Random Forest models, a soft voting model leveraging confidence intervals between their predictions was employed, as suggested by Ding et al. [1].

Furthermore, a K Nearest Neighbors (KNN) model was constructed to elucidate which biometric features were predictive of stress. Employing a 5-fold stratified cross-validation method on the entire dataset, encompassing classes from all four participants, facilitated a comprehensive exploration of feature contributions in stress prediction.

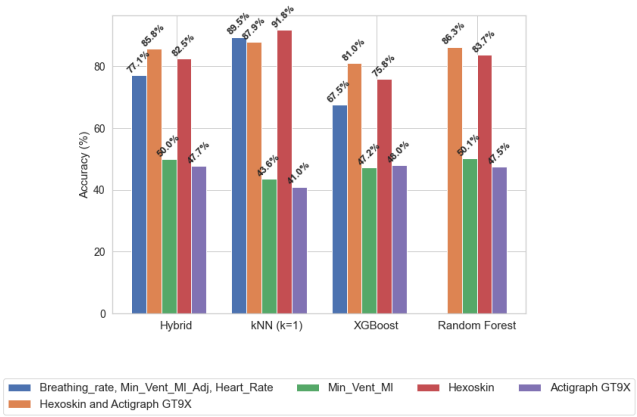

Figure 6: Model Performance – Feature Selection & Model Choice (Accuracy)

5 RESULTS

5.1 Model Predictions and Further Data Exploration

Initially, the performance of various models, including the Hybrid model, Random Forest model, XGBoost, and KNN Model, was evaluated across all features, Hexoskin and Actigraph. Notably, the KNN model exhibited superior performance, achieving an 87.9% accuracy within a runtime of one minute or less. (Figure 6)

To investigate potential differences in physiological responses to stress between STEM and non-STEM classes, an analysis encompassing all features from both Hexoskin and Actigraph was conducted. The feature importance order for STEM and non-STEM classes revealed negligible differences, as depicted in Figure 4. This observation was confirmed using the Hybrid model, averaging the feature importances from Random Forest and XGBoost. These findings underscore the absence of discernible distinctions in physiological stress response between STEM and non-STEM classes.

Subsequently, the predictive capacity of solely utilizing movement data from the watch to infer stress levels was explored. The XGBoost model yielded the highest accuracy among the four models, achieving a 48% accuracy, as illustrated in Figure 6. This outcome suggests that relying solely on movement data is not a viable approach for stress prediction.

The investigation then focused on determining whether the top feature, breathing volume, could serve as a reliable predictor of stress levels. Despite the absence of previous studies employing breathing volume for stress prediction, it emerged as a crucial contributor in our models. The Random Forest model exhibited the highest accuracy of 50.1% when exclusively utilizing breathing volume (Figure 6).

Recognizing the limitations of watch features as standalone stress predictors, attention shifted to assessing all Hexoskin features. The KNN model, incorporating all Hexoskin features, achieved the highest accuracy at 91.8%, surpassing studies employing more invasive equipment. (Figure 6).

Utilizing Shapley feature importance from the XGBoost model, the influence of specific features on different stress levels (0, 1, or 2) was examined (Figure 7). Although certain features exhibited preferences for predicting specific stress levels, breathing volume consistently emerged as the most robust predictor of the highest stress response.

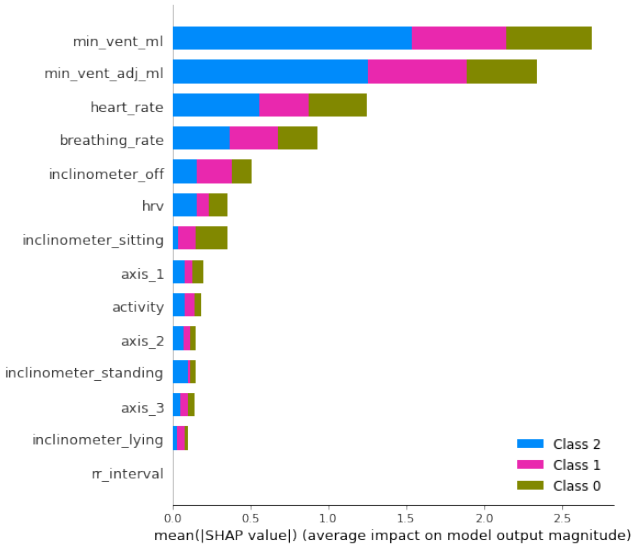

Figure 7: Shapley Feature Importance XGBoost

Following these results, features such as adjusted breathing volume, heart rate, and breathing rate were identified as pivotal for predicting stress (Figure 4). To simplify the model and comprehend feature importance, a pair plot was created, focusing on breathing volume, breathing rate, and heart rate. While this visualization provided decent separation of stress levels, a three-dimensional plot (Figure 5) offered a more comprehensive view.

The final evaluation involved running the four models exclusively on the top three features. The KNN model, with k = 1, exhibited the highest accuracy at 89.5%, approximately 2% lower than using all Hexoskin features. (Figure 6) Notably, the KNN model demonstrated superior speed, outperforming the Random Forest and Hybrid models by 3 to 4 times, with XGBoost being the fastest but exhibiting variable accuracy.

Adjusted Breathing Volume and breathing volume were identified as the most influential contributors to model performance, revealing a novel insight in the intersection of machine learning, sensor technology, and stress detection, even though the correlation between breathing volume and stress is acknowledged [14]. This significant accuracy achieved with only three features underscores the potential for unobtrusive stress detection in students, paving the way for the development of efficient stress-detecting sensors.

6 DISCUSSION

The exploration of stress detection in students represents a pivotal avenue of research, offering the potential for timely interventions to mitigate stress within a classroom setting. This research not only underscores the significance of alerting students to potential stress but also suggests practical strategies, such as deep breathing exercises, to alleviate stress. Although the need for a more extensive

participant pool is acknowledged to enhance the study's robustness, the examination of data from four participants over several months has yielded valuable insights. This approach allowed for the validation of findings against existing literature and facilitated the development of a more interpretable and computationally efficient model, particularly the K Nearest Neighbors model with a parameter setting of k = 1.

A noteworthy revelation from the study is the unexpected efficacy of utilizing movement data to predict stress with a high degree of accuracy. Additionally, the surprising enhancement of model performance with the inclusion of all Hexoskin features, beyond the initially identified top three features (breathing volume, heart rate, and breathing rate), underscores the novel nature of the insights gained into physiological responses to stress. The challenges encountered in this study were exacerbated by the use of unpopular and unique sensors, contributing to the unconventional nature of the data collection process. This uniqueness may have played a role in the unexpected identification of breathing volume as the most crucial feature for stress detection, a finding not commonly reported in existing literature.

Furthermore, the study grappled with issues such as sensor malfunctions and human errors in data usage, necessitating meticulous data cleaning. The novel and unconventional nature of the data collection process adds an extra layer of complexity to these challenges. The intermittent nature of stress incidents, where a student might experience stress only at the conclusion of a class, leading to an entire class being labeled with the maximum stress score, highlights the need for further refinement in stress labeling methodologies. Addressing this concern could involve employing smaller timeframes for stress labeling, perhaps intervals of 10-30 minutes, to provide a more granular representation of stress levels. While not a drawback per se, this aspect signifies an area of improvement and underscores the novelty of the study's approach, considering that prior studies have successfully employed similar time durations for stress labeling and associated activity [4].

7 FUTURE WORK AND APPLICATIONS

In future research endeavors, there exists a compelling opportunity to delve into more nuanced variations in physiological responses to stress among students and potentially different classes. Expanding the participant pool, extending the duration of data collection, and amassing a more extensive dataset would be essential for such investigations. At present, discerning deterministic patterns in an individual's response to stress remains a challenging task. Unraveling these intricacies may necessitate considerations of factors such as age, personality, and other individual characteristics, potentially requiring the development of additional machine learning models.

Furthermore, there is an interest in obtaining data from students experiencing stress at level 2. This could be achieved by introducing challenging and high-stress tests as an extension of the current study, allowing for controlled observations and a deeper understanding of stress responses. Additionally, exploring data collection at smaller time intervals is considered, as this approach holds the promise of providing more accurate insights into stress dynamics. Additionally, it could also be interesting to simply average the data for each class into one row and to create classification models on the averages of the data for each class.

It is noteworthy that the vest used in this study proved unobtrusive, resembling a shirt and thus offering a discreet means of monitoring physiological responses to stress. The inconspicuous nature of the vest, worn beneath regular clothing, ensured that participants were not burdened by the awareness of wearing additional monitoring equipment. This aspect contributes to the feasibility and acceptability of employing such unobtrusive devices in future applications.

In summary, future research could extend the current findings by exploring individualized stress responses, conducting controlled experiments at higher stress levels, and refining data collection methods to capture more detailed and accurate insights into stress dynamics. The unobtrusive nature of the monitoring device opens up possibilities for broader applications in real-world settings, such as educational environments, where stress detection could play a pivotal role in supporting student well-being.

8 CONCLUSION

Our investigation into physiological responses to stress has uncovered a complex and individualized landscape, emphasizing the uniqueness of stress responses among individuals. Notably, no discernible pattern in responses between STEM and non-STEM subjects emerged, highlighting the diverse nature of stress experiences across majors and class types.

One significant revelation from our research is the underappreciated role of breathing volume as a predictor of stress. Consistently identified as the feature with the highest importance when consolidating data from all participants, breathing volume emerges as a pivotal yet understated factor in predicting stress, providing novel insights not commonly found in existing literature.

Additionally, our study reveals the limited viability of wrist-worn watches as standalone predictors of stress. Despite initial expectations, the best-performing model using only movement data from the watch achieved a 48% accuracy, indicating the inadequacy of this approach for accurate stress prediction.

Furthermore, our research suggests that stress prediction can be remarkably simplified to just three features: heart rate, breathing rate, and breathing volume. Achieving an accuracy of approximately 89%, and slightly higher when considering all vest features, this finding holds considerable implications for stress detection using sensors. The prospect of developing a vest equipped with sensors measuring these three features provides an affordable and effective solution for students to detect signs of stress in a classroom setting. The unobtrusive nature of such a product, worn beneath regular clothing like a shirt, ensures minimal disruption to students' daily activities.

This stress-detection system, built on a larger dataset akin to studies such as [8] (albeit lacking breathing volume), could notify students of elevated stress levels in real-time. Leveraging the computationally efficient K Nearest Neighbors model with k = 1, students could receive timely prompts to engage in deep breathing exercises, potentially leading to reduced stress levels. This proactive

approach may not only contribute to improved academic performance but also support higher GPAs and enhance the retention of information.

In conclusion, our study not only contributes to the understanding of stress physiology but also underscores the practical implications for the development of unobtrusive and cost-effective stress-detection tools in educational environments. The prospect of a sensor-equipped vest measuring three key physiological features opens up avenues for real-time stress management, potentially positively impacting students' well-being and academic success.

ACKNOWLEDGMENTS

We would like to thank the data mining and connectivity lab of College of Charleston for providing the equipment needed to collect the data.

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